
Senior Linux Administrator · Senior Software Engineer · Systems Engineer · Database Administrator

Linux SME, hands-on team leader or mentor, documentation writer, and software developer of database and automation apps, with broad & deep knowledge that brings a holistic vision to projects.

Canadian  and USA  citizen, open to global opportunities.

Notable Achievements

Developed a database application and presented it at a trade show as a teenager.

Wrote and published software that was used by the US federal government, US Navy, city of Victoria BC, NJS&EA, fire departments, transportation companies, farms, and airports.

Started a dial-up ISP with only Linux servers in 1999, added wireless service in 2000.

Chosen by four telecoms to build their first Internet server network.

Security clearances in Canada and USA throughout career. Canada Secret clearance 2024-2034.

Skills Overview

Years Software and Computer Administration

- 20+ Software engineering including DB design
- 20+ Linux: Gentoo(20) Arch(5) Artix(4) SuSE(6) Debian(5) Ubuntu(10) Red Hat/RHEL(17) UNIX(2)
- 20+ Postfix, Sendmail, Courier, DNS, DHCP
- 20+ Network design, configuration, troubleshooting
- 20+ DB admin: MySQL MariaDB PostgreSQL Sybase
- 20+ Web apps: Drupal WordPress PHP LAMP
- 20+ Security: firewall design, admin, troubleshooting
- 17 Version control: SVN Git CVS TFS Perforce
- 20+ Automation scripting: BASH KSH Perl PHP
- 12 Network monitoring. SevOne Zabbix Nagios SCOM
- 8 Virtual environments: VMware Proxmox Azure
- 20+ Storage tech: NFS Gluster ZFS RAID LVM BTRFS
- 2 Automation tools: Puppet Ansible

Business and communication skills

- Management of local and global teams
- Project and department management with budgets up to \$3M.
- Documentation: policies, procedures, product manuals, technical manuals, training, wikis.
- Teaching, mentoring
- Management of global remote teams
- Security audits
- Development of best practices
- English mother tongue; basic Italian & French

Professional History / Portfolio

ISP Ltd., New Glasgow, NS Canada <https://ispltd.com>

4/1999–present

President, Director

PHP, Perl, BASH	Apache, Sendmail, Postfix SMTP, Courier and
Proxmox Virtual Environment	Dovecot IMAP, BIND DNS, DHCP server,
Linux: Gentoo, RHEL 7-9, Oracle 7-9,	MySQL, MariaDB, PostgreSQL, Squid
Rocky 8-9, Debian, Ubuntu, Arch, Artix	GitLab, Dokuwiki, Mediawiki, SonarQube,
Puppet Ansible Cfengine custom scripts	Roundcube, NextCloud, Drupal, WordPress
Zabbix	TCP/IP networking: subnet, routing, firewalls
Server design, building, and maintenance	

Family business through which the contract roles below have been done.

Linux system administration, automation, configuration, management, web development.

Published a pioneering Linux ISP tutorial in 1999 that was used by educators.

Creation, hosting, and management of about a half-dozen *pro bono* web sites for charities. Firewall design and management. Database administration, software engineering, upgrades, monitoring, etc.

Published proprietary and open source software including *Stapler* DNS manager, *efind* Gentoo package manager enhancement, *File Retention Manager* storage manager, WVH vehicle maintenance tracker, WJS job tracker, and WML mailing list manager. Authored and published Linux administration tutorials, references, and automation scripts.

Building & maintainance of desktops and servers. Orchestrated data centre & platform migrations.

Implemented a Proxmox virtual environment cluster with five nodes and a backup server. Storage technologies include NVMe, SSD, HDD, ZFS, and Linux MD RAID. As of June 2024 the cluster has about 200 VMs and containers (LxC) with these Linux distributions:

- Red Hat Enterprise Linux 7, 8, and 9
- Oracle Linux 7, 8, and 9
- Rocky Linux 8 and 9
- Gentoo Linux
- Arch and Artix Linux
- Debian Linux
- Ubuntu Linux
- Solaris

Automation with in-house developed tools, Ansible inventory and playbooks, and Puppet.

Installation, monitoring, and management of:

- DNS servers (internal and external/public-facing)
- NFS servers
- DHCP servers (fault-tolerant)
- Web servers running WordPress, Drupal, and custom PHP apps
- Nginx cache and proxy servers
- Squid proxy servers (caching and filtering)
- Postfix mail servers
- Proxmox Mail Gateway
- MariaDB / MySQL database servers
- PostgreSQL database servers
- Streaming media servers (icecast and Rocket)
- Security software (Fail2ban, iptables firewalls, Proxmox firewalls, custom intrusion detection and reporting software written in-house)

PHP, SQL, HTML, CSS, JavaScript coding	Server and application security & monitoring
BASH and Perl scripting	Linux server administration and updates
MySQL & MariaDB config, admin, replication	
Gentoo and Arch Linux administration	Communication, verbal and written
Apache, Nginx, Sendmail, Postfix, DNS	Documentation writing
Subversion administration and use	Customer support

Software architecture & engineering, change execution of multiple LAMP web database applications serving the oil and gas industry. Team manager for initial release in 2002. Project manager of Android application R&D 2020-2022.

Gentoo and Arch Linux server administration (VMs and bare metal). MariaDB (MySQL) database administration and replication. Security audits, server monitoring, firewall management, end-user technical support, &c.

Planning and execution of data centre moves and software upgrades and migrations.

- Capacity planning
- Hardware and software selection
- Hardware and Linux software upgrades and replacements
- Data centre migration
- Server monitoring
- Process automation
- Server security
- Database upgrades
- On-site and remote back-ups

Notable achievements:

- Initial release finished c.10% ahead of schedule and under budget.
- Designed intrusion detection, reporting, and mitigation scripts.
- Geographically distributed servers that provided continuous availability during maintenance periods or network outages.
- Migrated from PHP 3 to 5, 7, and 8 without any down-time.
- Migrated from MySQL 4 to 5 to MariaDB 10 without any down-time.

Linux server administration	Kubernetes and Docker configuration
MySQL, MariaDB, PostgreSQL	Upgrade from Puppet 6 to 8
Red Hat Linux 7 and 9 administration	Git and GitLab usage and administration
Upgrade from RHEL 7 to 9	Azure VMs and PaaS
Linux patching, monitoring, troubleshooting	Configure Azure AD authentication in Linux
Server automation & monitoring scripts	Documentation writing
Puppet administration, configuration, scripting	Mentoring

Worked with Canada's PSPC providing operation management and support of the Canada Buys web site infrastructure — production and non-production environments.

Red Hat Enterprise Linux server administration. Server patching / upgrading. Issue diagnostics and resolution. Server performance monitoring and tuning. Evaluation, planning, upgrading RHEL 7 to versions 8 and 9. Mentoring. Documentation. Apache, Nginx, Solr, Zookeeper, GitLab server, Puppet, PuppetDB, PostgreSQL, MySQL.

Development of procedures and proofs of concepts for SonarQube, Livezilla, GitLab, and others using VMs in a self-managed ProxMox virtual environment in my personal lab as well as in Azure cloud VMs and docker containers.

Lead administrator for Puppet configuration management. Developed and implemented procedure for managing multiple versions of PHP on Red Hat servers, and for migrating sites to the newer version without down-time.

Troubleshooting of MySQL and MariaDB issues. Maintenance and troubleshooting of Azure tooling and Azure Active Directory back-end authentication issues on Linux servers.

Monitoring of server performance and rapid response—from identification to solution and implementation—for Production issues.

BASH scripting for automation	Patch procedures development and team guidance
Oracle & Red Hat Linux 6, 7, 8 administration	Documentation: policies and procedures
Install & configure Apache Wiki Postfix DNS	Security, configuration, and performance audits
VMware	Training/mentoring

Linux software engineering, server administration, troubleshooting, solution designs, documentation, security audits, server automation programming, development of custom server monitoring software to detect complex anomalies, development of and upgrades to VMware server templates, Identity Management configuration, NFS, Active Directory integration.

Red Hat Enterprise Linux and Oracle Linux.

Linux administration tasks included:

- Administration, configuration, troubleshooting Oracle Linux and RHEL versions 6, 7, and 8.
- Designing, writing, and implementing a flexible job scheduling solution that worked across an arbitrary number of servers, scheduling jobs based on time and other job dependencies.
- Designing, writing, and implementing a network-wide service for managing data file retention, archiving, and clean-up.
- Designing and implementing of a method of efficiently archiving and transferring high volumes of data (millions of files, terabytes of data).
- Consulting on best practices, security, automation, and deployment of Linux for global (parent corporation) ERP system, Provincial power grid controllers, and Provincial smart meter initiative.
- Server performance and security audits and automation.
- Writing and delivering reports for department management.
- Server changes for improved stability, security, and performance.
- Process and monitoring automation development and implementation, including intelligent network file system monitoring software with features not attainable in Oracle OEM.
- Server template development and usage.
- Identity management
- Linux authentication using Windows Active Directory
- Kerberos key distribution centre (KDC) configuration
- Developing and implementing a technical knowledge base, local software mirror, and ERP (global) patching procedures, and best practices documentation.
- Patching strategy and documentation for critical Linux servers involved in electrical transmission and distribution control and monitoring (SCADA).
- Administration and documentation of Oracle and Red Hat Linux servers in Dev, UAT, and Production, both stand-alone and virtual servers in a VMware environment.

SevOne & Spectrum NMS administration	Database administration: MySQL, Sybase
Red Hat Linux 4, 5, and 6	Spectracom GPS clock config. and admin.
Gentoo Linux server administration	Technical documentation
SevOne PHP mods & enhancements	JIRA, Twiki, MS Office
Perforce and Git version control	Vendor relations, communications, support
BASH KSH Perl	Reporting to management
SNMP NetFlow LDAP NTP	Technical support
Configuration automation scripting	

Systems and software engineer for telecom team. Engineered tools for monitoring, managing, and automating the administration of over 16,000 core routers, switches, and servers of a global financial network as part of the telco engineering team.

Angelo's work ethic and expertise are fabulous.

—Rudy Deca, Network Engineer, Morgan Stanley

Managed hundreds of globally-distributed and clustered servers running Red Hat Enterprise Linux (RHEL 4, RHEL 5, and RHEL 6), Gentoo Linux, and a small number of Windows servers.

Developed Kerberos KDC configurations.

Managed globally-distributed SevOne NMS cluster, Spectrum NMS, and in-house developed Perl software for monitoring and automation.

Architect and engineer of solutions for client needs using PHP, Perl, BASH, and SQL.

Authored technical and end-user documentation for our group's products.

Authored server automation policies, procedures, scripts, and documentation.

Contributed to the enterprise's global documentation reference library.

First Advantage, St. Petersburg, Florida USA <https://www.fadv.com>
Software Engineer IV, Technical Team Lead

2/2012—11/2012

PHP and SQL coding	Team management
MySQL DBA	Policy development
RHEL / CentOS 6 Linux administration	Reporting to management
TFS (version control) use & policies	Technical documentation

Software architect and manager of geographically distributed team of 13 that were located in the USA and overseas. Florida office was moved to Indiana at the end of the year.

Red Hat / CentOS Linux 6 server administration. MySQL database administration.

Author of departmental policies and standards for software and server architecture.

Security audits, procedure writing, change implementation.

Director and mentor to Linux SA and DBA. Acting Linux SA and DBA *pro tempore* for three months.

J.D. Irving, Saint John, NB
Senior Red Hat Linux Consultant

2/2010—12/2010

Red Hat Enterprise Linux (RHEL)	Business requirements analysis
Perl, BASH, awk, sed	Product research, reports, recommendations
Automation	Documentation

Provided definition, analysis, and implementation advice and documentation on new Linux projects.

Red Hat Enterprise Linux (RHEL 6) server administration for ~110 servers in the areas of:

- Automated VM deployment with Kickstart
- Server & network infrastructure monitoring with Zabbix
- Configuration management with Cfengine
- Software management with Cfengine and Subversion
- Linux integration with Active Directory, Kerberos KDC configuration

J.J. MacKay Canada Limited, New Glasgow, NS

9/2008–5/2009

Head of IT Department, Senior Systems Analyst, Senior Application Developer

Technical Skills & Roles

Windows 2003 server administration
Gentoo Linux server administration
RHEL & CentOS server administration
PHP SQL HTML CSS JavaScript BASH Perl
Software architecture
MS SQL Server 2005 administration
MySQL and PostgreSQL administration
Apache web server (Windows & Linux)
IIS 6 & 7 administration
CVS (version control)

Administrative Skills & Roles

Department management
IT support services
Project management
Documentation, knowledge-base wiki
Security audits and policies
Software licence audits and policies
Network planning, monitoring, and alerting
Corporate desktop software standards

Director of a small IT department. Responsible for corporate network, servers, and desktops. Managed 13 projects, internal & out-sourced personnel, vendor relations, etc. Implemented changes in software architecture that significantly reduced licence costs and development effort.

CentOS (Red Hat) Linux server administration.

Contributed to PCI DSS compliance certification: Wrote policies and procedures, recommended and implemented security-hardened Linux servers, and converted a C++ utility from Windows to Linux. A Trustwave security audit found zero issues on the first pass. Led development of two web-based desktop+mobile database projects. Improved SQL features and efficiency, resulting in report generation speed improvements of approx. 45-times (i.e., from minutes to seconds).

Morgan Stanley, Halifax, NS

8/2007–9/2008

Senior Consultant

RHEL Red Hat Linux 4 & 5 administration
Windows 2000 & 2003 Server administration
Sun Solaris 8 administration
Apache 1.3 / Tomcat 5.0 & 5.5 administration
IIS 5 & 6 administration
Proxy server administration: iMimic, Inktomi
Perl, BASH, KSH, PHP, RCS, SyBase
Java application troubleshooting
SiteMinder administration, Kerberos
Cisco & F5 load balancer administration

L2/L3 support, night-shift lead
Technical documentation writing & editing
Monitoring of high availability global network
Incidence reporting & teleconference facilitation
Multi-tasking communicating via phone, IM, e-mail, and in-person, frequently all four simultaneously
SharePoint database administration
TWiki support & maintenance
Netcool / Falcon enterprise monitoring / ticketing

Member of the team that was responsible for monitoring and maintenance of infrastructure serving high-profile private banking customers and global stock markets.

Responsible for the troubleshooting and administration of over a thousand Red Hat Enterprise Linux (RHEL 4 and RHEL 5) servers and just over one hundred Solaris and Windows servers.

Wrote documentation & scripts that were adopted as enterprise standards.

SAWTNET, Oran, Algeria**1/2004-12/2004**

Linux Systems Engineer (contract)

Responsible for architecture, building, and exporting of a turnkey Linux server network for this telecom to provide Internet services via dial-up and wireless.

Work was done remotely from Victoria, BC, Canada. Physical servers and network equipment were exported to Algeria after all programming, integration, and testing was completed.

Monarch Communications Limited, Lagos, Nigeria**10/2002-1/2004**

Project Manager, Systems Engineer (contract)

Architect of a Linux server network for 100,000 customers. Led team of 11 people. Asked to complete about 40 days ahead of schedule and succeeded in doing so.

Work was done in Victoria BC Canada and Gaithersburg MD USA. Physical servers and network equipment were exported to Nigeria after programming, integration, testing, and training was completed.

Access Point Communication, Port Harcourt, Nigeria**11/2002-5/2003**

Project Manager, Systems Engineer (contract)

Architect of a compact ISP server network: only 12U including UPS. In-person and remote training (over Internet). Guided remote reconstruction of a server.

Work was done remotely from Victoria, BC, Canada. Physical servers and network equipment were exported to Nigeria after all programming, integration, and testing was completed.

M&V Commsat Ltd., Luanda, Angola, Africa**1/2001-12/2002**

Project Manager, Systems Engineer (contract)

Network architect. Managed manufacturing, assembly, exporting. Led 18-day installation & implementation team in Angola. Original servers were still operational after seven years and M&V Commsat had grown to be one of Luanda's largest and most stable ISPs.

Network was manufactured in Paonia, Colorado then exported to Angola, after which I went with a team of three on-site for three weeks to complete the installation.

Paonia.com (Internet service), Paonia, Colorado USA**1/1999-6/2002**

Owner, Systems Engineer, Linux System Administrator

Launched an Internet service that ran entirely on Linux servers. Research, implementation, marketing, operation. Developed wireless broadband infrastructure that spanned 12 miles.

Developed family-oriented filtered web browsing service.

Education

Drake University, Des Moines, IA. Nat'l Science Foundation honour course in physics & computers.

University of Minnesota, Institute of Technology, Minneapolis, MN. First year (30 credits) of BSEE.

Certifications

2007 Dalhousie University, Halifax, Nova Scotia: Perl Programming course

2008 Dalhousie University, Halifax, Nova Scotia: Software Project Management course

2008 Business Objects Crystal Reports, level I, II, and III certification

2008 Hands-on Tech., Columbia, MD: MS SQL Server 2005 Administration certification
2008 Hands-on Tech., Raleigh, NC: C# and ASP.NET certification
2010 Zabbix SIA at UniRede, Porto Alegre, Brasil: Zabbix Certified Specialist
2010 Zabbix SIA at UniRede, Porto Alegre, Brasil: Zabbix for Large Environments certification

Spoken Languages

English (fluent, mother tongue) • French (basic vocabulary) • Italian (basic vocabulary)

Volunteering and Hobbies

Volunteer broadcast engineer. Produced musical, news, and original comedy & variety shows.
Volunteer and paid audio engineer, live and studio work with 6- to 40-channel mixing consoles.
Volunteer web site design and hosting for charities.
Teacher and preacher of the Holy Bible on-line and in-person.
Music. Travel. Languages. Ancient Greek studies (2018-2020 St. FX University, Nova Scotia).

Web Site References

Sites designed for clients

<https://donsdirectory.com> — Oil & gas industry subscription database web-app. Linux, Apache, Nginx, HTML, PHP, CSS, Javascript, and MariaDB / MySQL replication. No frameworks; 100% original design. Also designed a back-office web-app that manages the data and has accounting features (not open to the public, but I can give a demo of the back office site).
<https://ashkellysroofing.com> — Hosting and development of a Drupal site for a local roofing contractor in New Glasgow, NS.
<https://bible.mikeattwood.org> — video hosting platform. Server and site software selection, installation, configuration, and maintenance. Linux, Apache, PHP, MariaDB.
<https://jah.ca> — audio hosting site. Linux Apache MariaDB PHP. All original work, designed for a church in 2004, continually updated since then as a personal project.
<https://patientrm.com> — WordPress site. I did not do the initial design, but took over the management and further design work.
<https://altenberg-ib.com> — WordPress site. I did not do the initial design, but took over the management and further design work.
<https://hjperumission.org> — WordPress site. Configured site, chose and configured plug-ins, designed the look from content provided by client.

Sites promoting my own work

<https://ispltd.com> — Linux consulting and software development. Linux, Nginx, PHP, MariaDB, all original design work.
<https://ispltd.org> — Linux tutorials, tips, and notes for system administrators driven by DokuWiki (Linux, Nginx, PHP).
<https://ispltd.net> — Linux server management, driven by WordPress (Linux, Apache, MariaDB, PHP).